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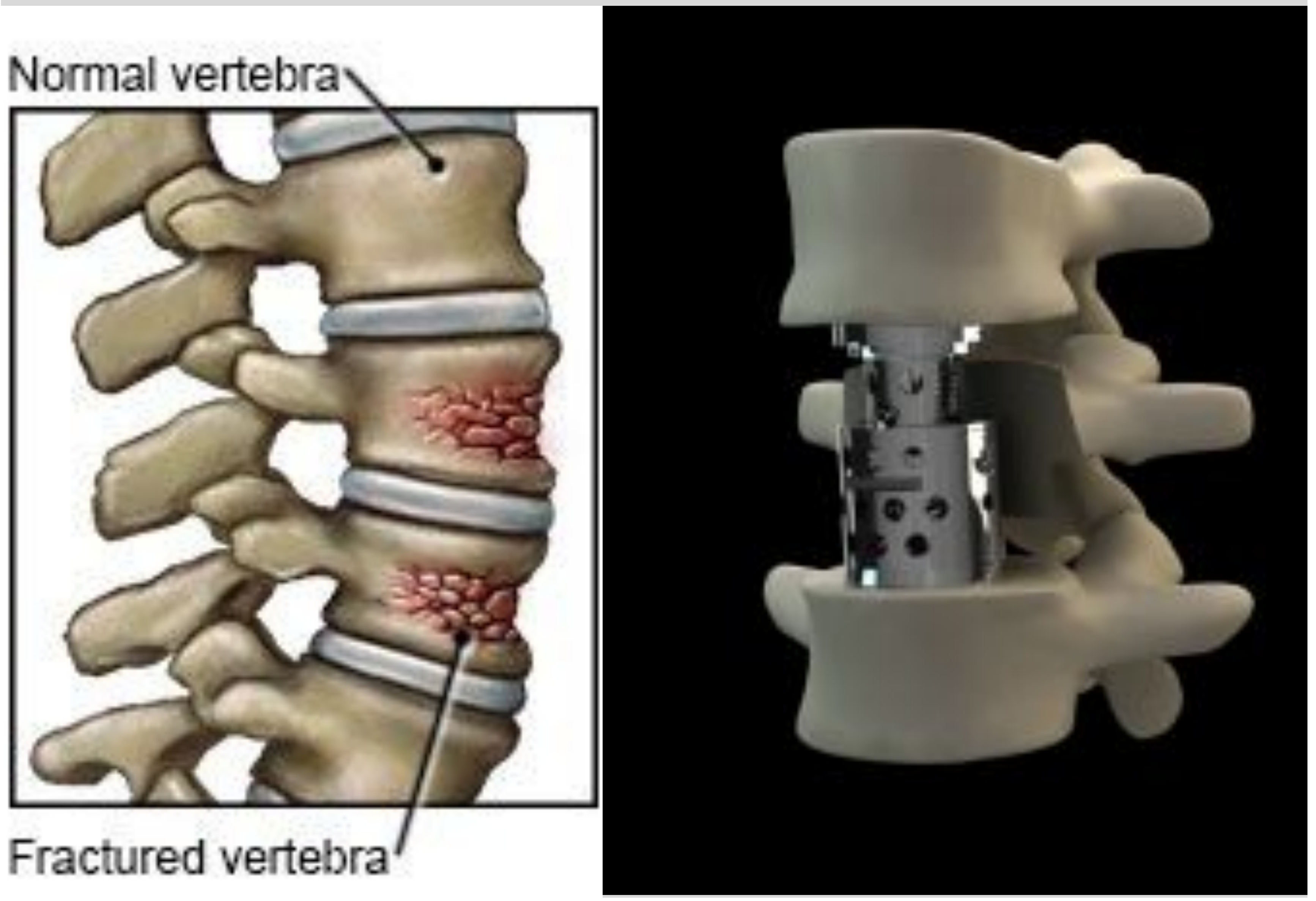
ENGN 1490
Biomaterials

# Next Generation Material Selection for Vertebral Body Replacement (VBR)

**Compression Fracture in the Spine (Left) and the expandable titanium GIZA™ VBR system (right)**




Osteoporosis, fractures, tumors, and infections in the spine severely compromise its mechanical stability and cause severe pain and physical immobility. Ideally, VBR devices aim to restore the function of the diseased vertebrae, maintaining appropriate stability and mobility of the spine.

| Current | Future |
|---|---|
| - Titanium<br>- PEEK | - CFR PEEK<br>- Porous Tantalum<br>- Biodegradable Magnesium |

## Material Selection Criteria

| | |
|---|---|
| Young's Modulus | Percent Elongation |
| Compressive Strength | Durability |
| Fracture Toughness | Sterilizability |

## Initial Material Selection

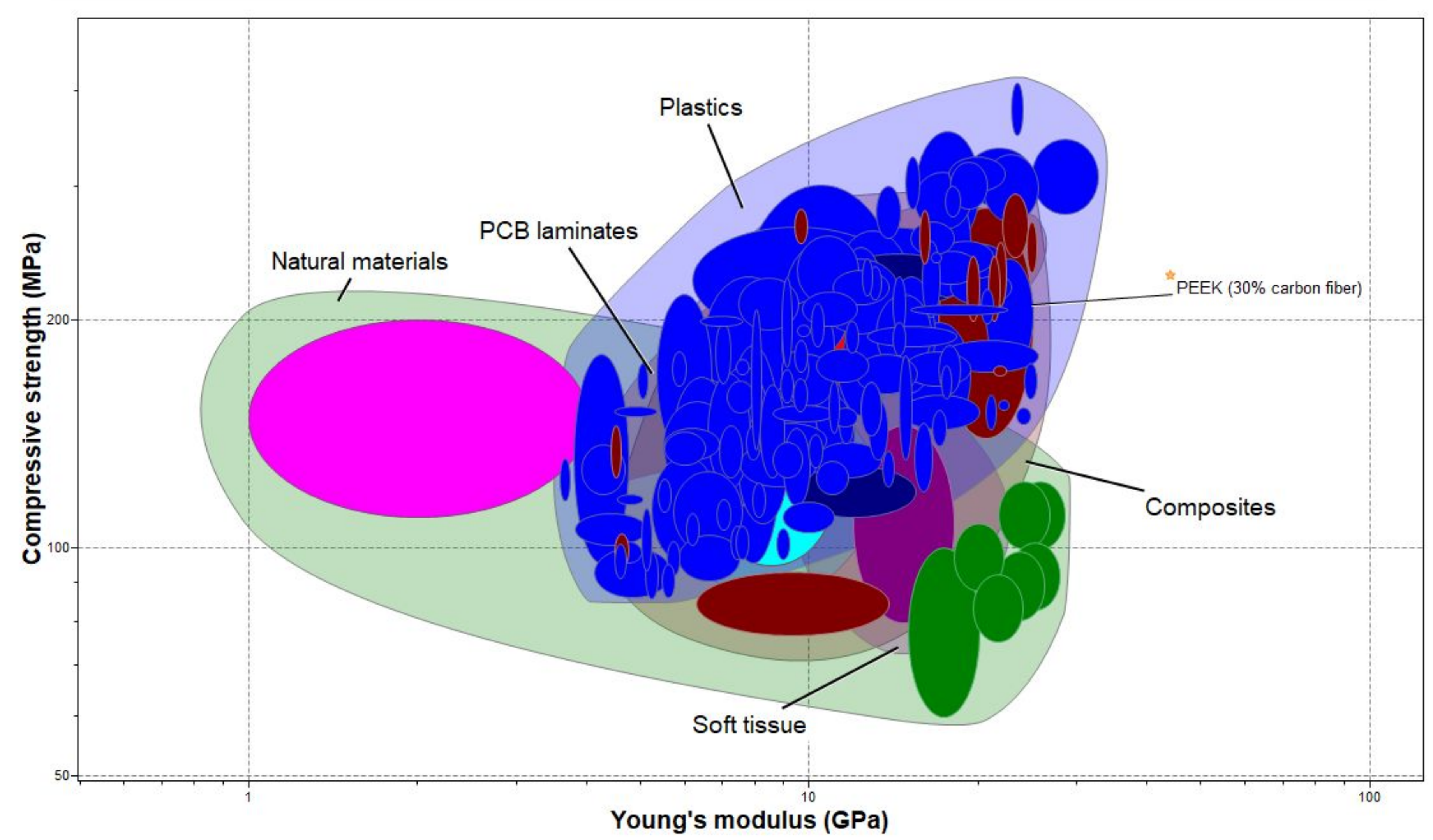



**Figure 1**. Young's modulus (GPa) vs. Compressive Strength (MPa) after the initial material selection stages with material envelopes labeled in addition to keratin in pink and antler in light blue.

## Material Refinement

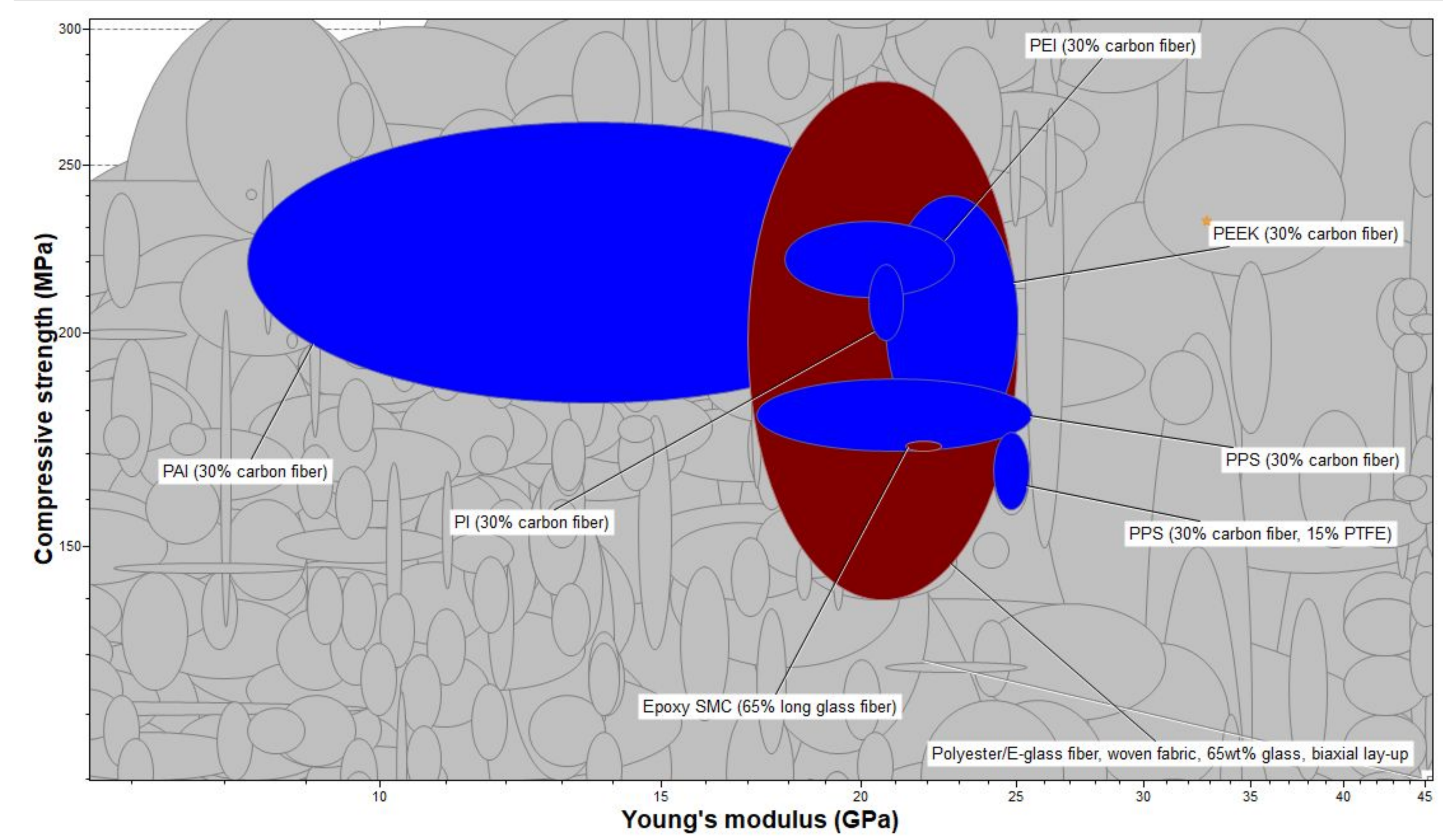
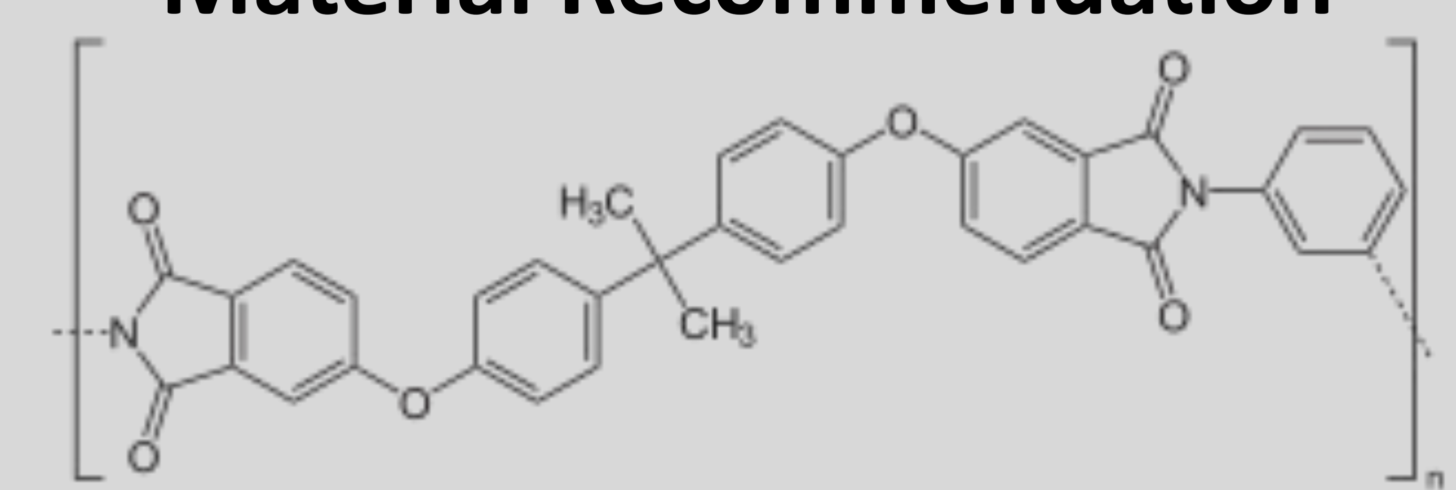



**Figure 2**. Young's modulus (GPa) vs. Compressive Strength (MPa) after the final refinement. Passing composites are shown in maroon and passing plastics are shown in royal blue.

## Material Recommendation



Polyetherimide (PEI)

| | |
|---|---|
| Suitable mechanical properties | Biocompatible |
| FDA-approved for bone fixation systems | Cost-effective |

Acknowledgements: Thanks to Dr. Coulombe for guidance on material refinement
Figures from ANSYS, Inc. Ansys Granta EduPack Software. 2025 R2; version 25.2.1